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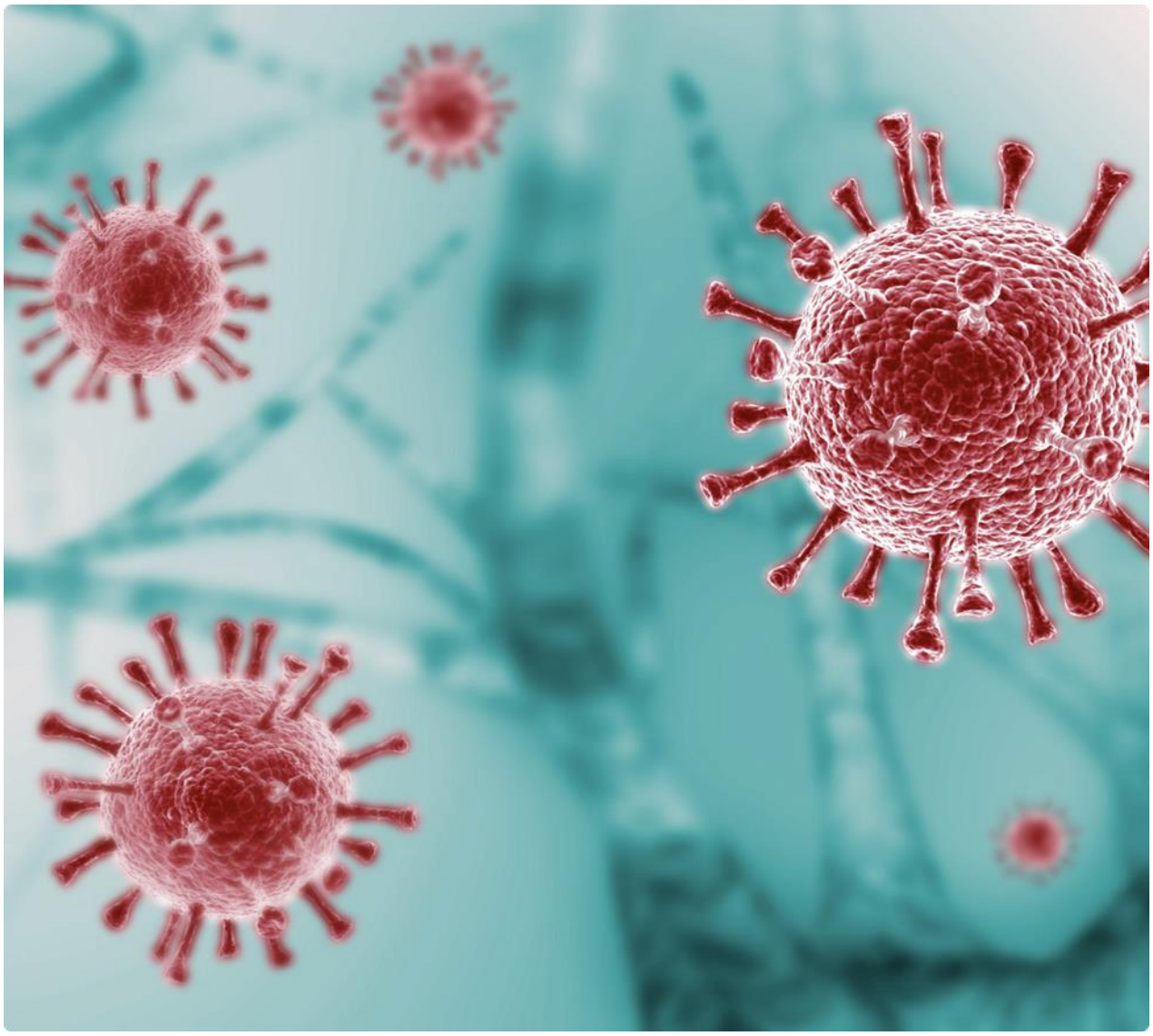
Analysis of COVID-19 Spread and Impact Using Geographic and Demographic Data



Introduction and Background

The COVID-19 Pandemic has reshaped global health dynamics, embodying a dire need for integrative data to describe the virus and its dynamics pertaining to transmission. Our platform designs a robust database model against which multi-dimensional associations existing between virus transmission, people's demography, population healthcare infrastructure, and mass transportation networks could be analyzed consistently. In fact, due care in this regard as an integrated approach is able to bring into light every pattern of COVID-19 spread, as well as response strategies and their effectiveness when applied across different regions.

By harnessing data from population statistics, healthcare facility records, and airport traffic records, we attempt to create an insightfully rich picture of virus dissemination and the impact on heterogeneous population subgroups. Our model goes further to capture, aside from the direct interaction with the virus, some general socio-economic implications using coupled patterns of mobility with accessibility to healthcare.

This research aims to enhance our understanding of the dynamics of the pandemic with spatial data analysis and informs decisions on public health with respect to critical areas for intervention and resource allocation. Mapping such complex interactions of data in this project gives insight that is of great importance for enabling policymakers and healthcare providers to be better prepared to tackle the ongoing challenges brought by COVID-19 and all future public health crises.

Research Questions and Objectives

Our study addresses several key questions in trying to dissect the breadth and depth of the pandemic from many different angles:

• How do population density and mobility patterns affect the rates of transmission of COVID-19 in urban and rural settings?

• How does COVID-19 relate to the outcomes based on the availability of facilities and accessibility to hospitals?

• Does the proximity to major transport hubs affect regional COVID-19 transmission dynamics?

Database

We used the U.S. Census Bureau to provide detailed demographic data on how population variables affect the spread of COVID-19. We also obtained data on healthcare facilities from the Department of Health and Human Services, which included the number of hospitals and whether they provide emergency services, critical for assessing medical accessibility and response capabilities in various areas. The authors made use of airport traffic data from the Federal Aviation Administration to illustrate how major transportation hubs became a critical factor in regional COVID-19 dynamics.

Taking into consideration the diversity of the sources of data, quite heavy preprocessing was performed for achieving data consistency and usability and accuracy.

First, we cleaned the data rigorously to iron out inconsistencies in the format of data representation and missing values. For example, county names were standardized to match across different datasets. We then merged these datasets on common identifiers such as state and county names for integrated analysis.

We normalized the data to compare it across different scales. This was especially true of population and infection rate datasets. Geospatially enriching the datasets further allows for spatial analysis on the datasets. Latitude and longitude coordinates were added for the healthcare and population data points, thus enabling mapping both resources and demographic factors to be mapped with precision pertaining to infection rates. Lastly, the year 2021 was selected because it is an important year that signifies one full cycle of COVID-19 impact. This will enable us to capture the progress and responses of the pandemic in the varying regions comprehensively. Besides, the data for the year 2021 is relatively mature and complete; thus, increasing the reliability of the analysis.

Conceptual Design

This entity-relationship diagram emphasizes the interlinked aspects of data about the COVID-19 pandemic, including the spread of the virus, its effects on populations, the role of transportation infrastructure, and the services provided by healthcare facilities. It offers an integrated view of how the pandemic interacts with different systems and demographics.

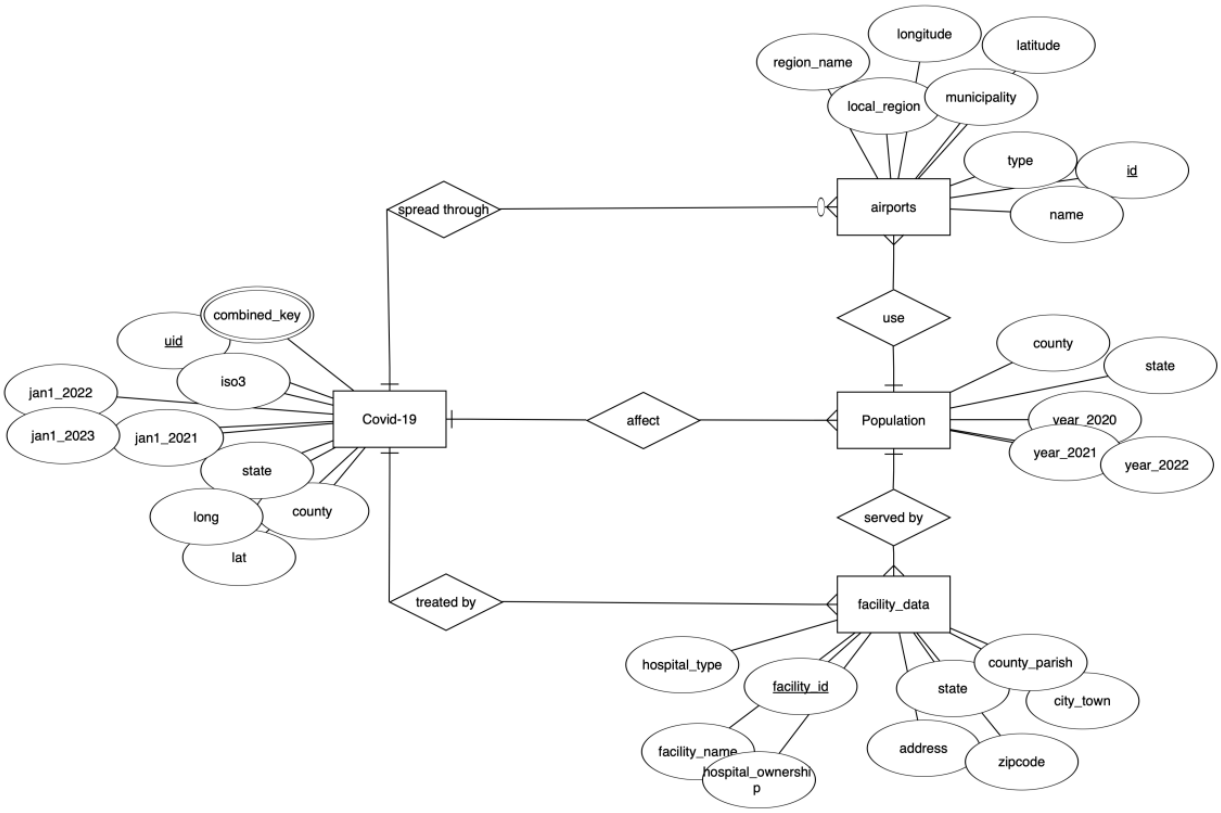


Figure 1: Spatial Database Conceptual Design for COVID-19 Data Integration

Logical Design

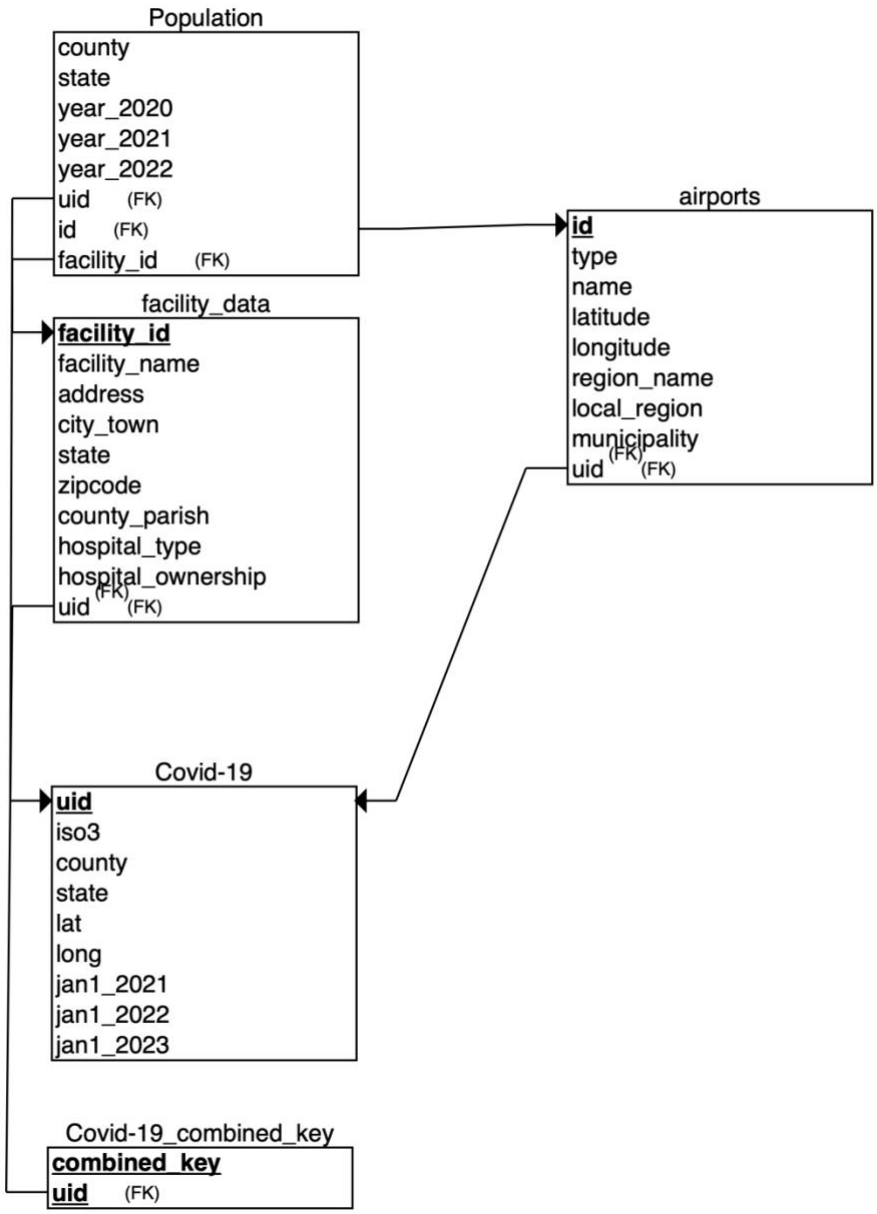


Figure 2: COVID-19 Data Integration Logic Diagram

We develop a set of normalized relations that implement the conceptual design represented by the diagram, in which each of the major entity types-population data, facility data, airport, and Covid-19-related data-is assigned with a surrogate primary key unique to identify individual records within the dataset. The field for natural keys for respective entities coincides with definite dimensions that refer to a set of additional supporting entity types, which ensure referential integrity.

Methods

The SQL query will be used to extract COVID-19 infection data and calculate the infection rate for 2021 from a database for counties across the United States. The results are meant to be used in creating visual representations on the ArcGIS Online platform. This visualization can efficiently depict the severity of the pandemic across different regions.

```
SELECT
        c.county,
        c.state,
        c.jan1_2022,
        p.year_2021,
        ROUND((CAST(c.jan1_2022 AS NUMERIC) / NULLIF(p.year_2021, 0)), 3)
        AS Infection_rate_2021,
        c.lat,
        c.long
FROM
        covid19_data c
JOIN
        population p
ON
        c.county = p.county AND c.state = TRIM(p.state)
WHERE
        (CAST(c.jan1_2022 AS NUMERIC) / NULLIF(p.year_2021, 0)) > 0 AND
        (CAST(c.jan1_2022 AS NUMERIC) / NULLIF(p.year_2021, 0)) < 1
GROUP BY
        c.county,
        c.state,
        p.year_2021,
        p.year_2022,
        c.lat,
        c.long,
        c.jan1_2022,
        p.year_2021
ORDER BY Infection_rate_2021 DESC
```

	county character varying (50)	state character varying (50)	jan1_2022 integer	year_2021 integer	infection_rate_2021 numeric
1	Chattahoochee	Georgia	6299	9044	0.696
2	Loving	Texas	27	54	0.500
3	Dimmit	Texas	4022	8479	0.474
4	Crowley	Colorado	2525	5735	0.440
5	Dewey	South Dakota	2224	5223	0.426

Figure 3: Top Five U.S. Counties with the Highest COVID-19 Infection Rates in 2021

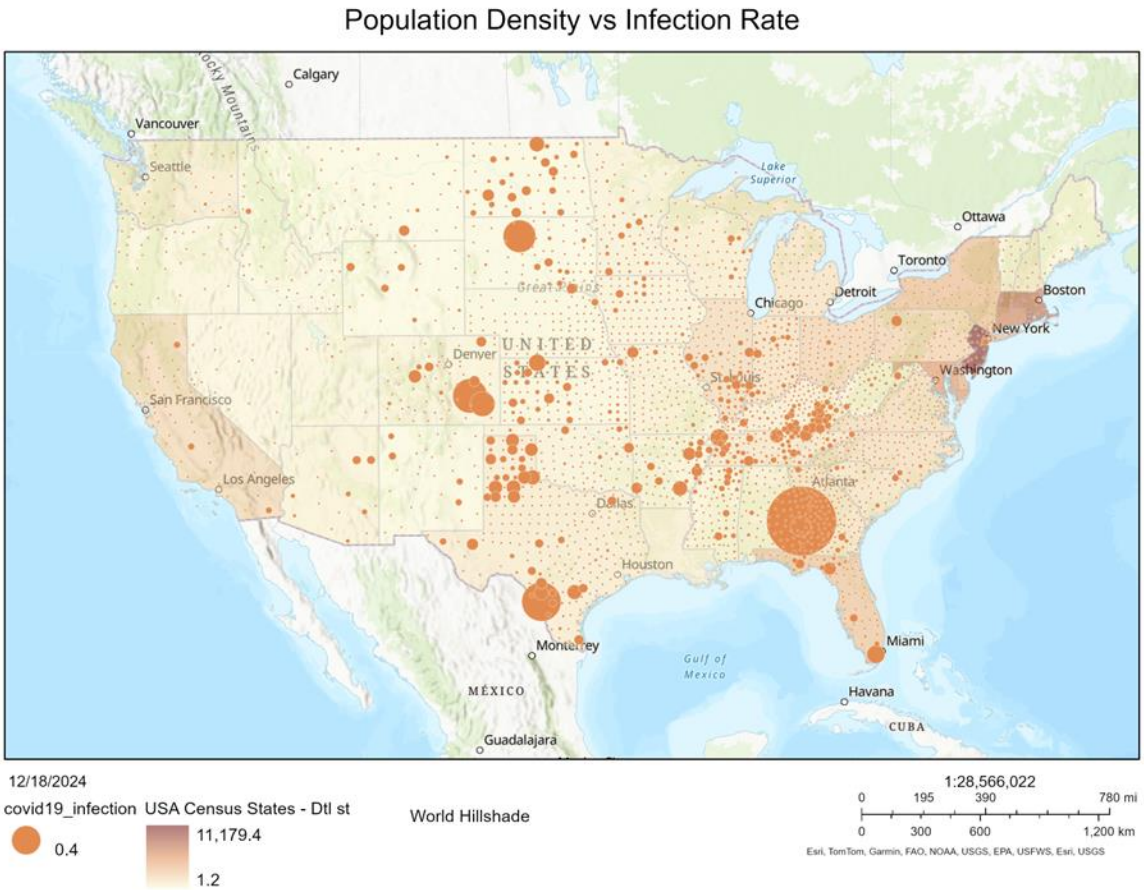


Figure 4: Spatial Distribution of COVID-19 Infection Rates Relative to Population Density in the U.S.

There are two visualizations: the one on the left showcases five counties with the highest rate of COVID-19, and the second was a national map created by me in ArcGIS Online showcasing Infection Rates across U.S. counties. These above counties appear as some large dot markers on the National level map and are largely spread along the central United States as shown in the image on right. Intriguingly, these markers are missing in the most heavily settled areas, represented here as darker shades on this map. Such an observation indeed suggests that high population density does not correlate directly with high infection rates but instead that disproportionately high infection rates have shown up in such sparsely populated rural areas.

It is probable that the infection rates in these counties that have been observed are probably related to several major factors.

1. **Inadequate Medical Facilities:** The shortage of adequate medical facilities and professional health personnel in rural areas is often the result of the far-flung nature of these communities and the low economic activities therein.

2. **Low Population Mobility:** Although the mobility is generally low in rural areas, it may also result in the quicker transmission of the virus within the community once the virus has penetrated the community. In a small close-knit community where interpersonal contacts are quite frequent, the virus may spread fast and undetected once an outbreak has begun.

In the light of these suppositions, it is essential to delve into the medical facilities and transportation infrastructure, which are the airports in such areas. This will help in identifying whether it is, in fact, limited health resources and mobility patterns that mainly contribute to the rates that are observed as high beyond what could be described normally in less densely populated rural areas. This might also give better insight into what dynamics are involved, thus possibly leading to more effective public health interventions and resource allocation.

Medical level and Infection

SQL query:

```
SELECT
   s.state_name,
   COUNT(f.facility_id) AS facility_count
FROM
   facility_data f
JOIN
   us_states s ON f.state = UPPER(s.state_abbreviation)
GROUP BY
   s.state_name;
```

The SQL query below counts all facilities in each U.S. state by joining in some facility data to this state data. This results in a summary of facility counts per state that can then be analyzed and visualized in tools such as ArcGIS online.

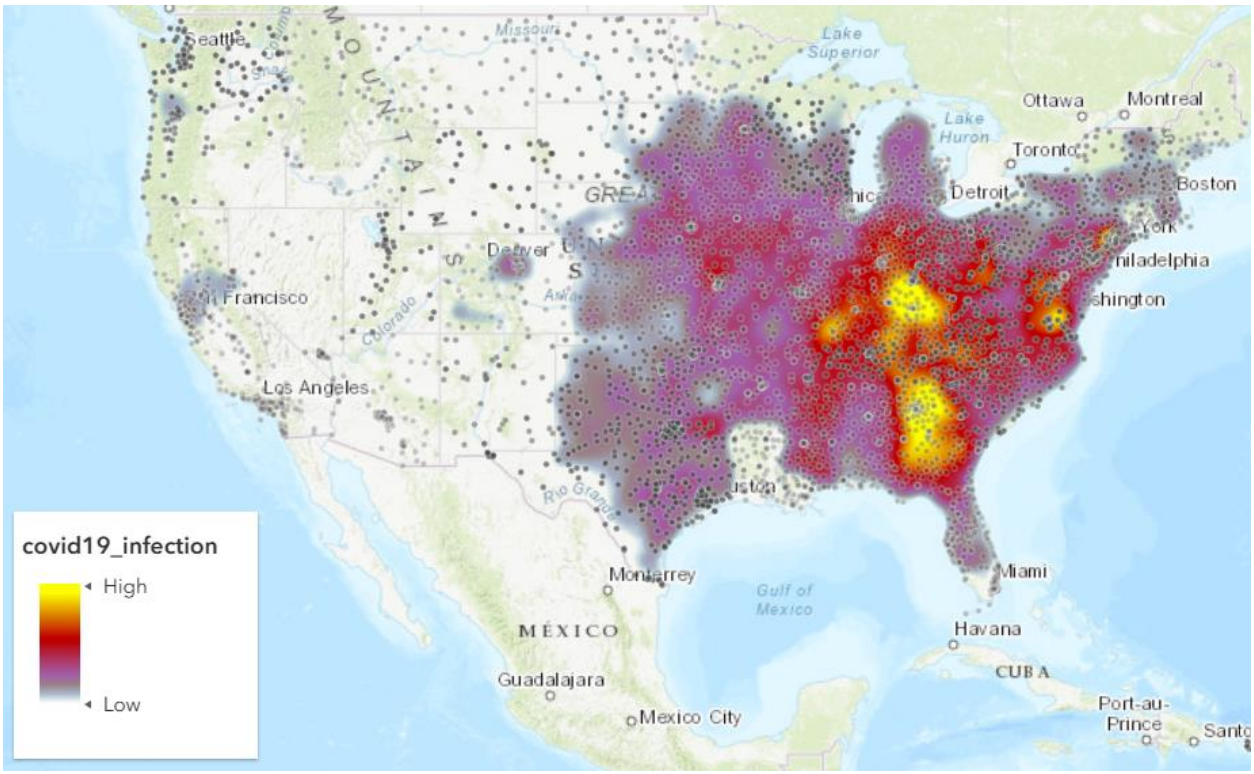


Figure 5: The Geographic Spread of COVID-19 Infections in the U.S.

Below is the heatmap of infection rates for COVID-19 across the United States, combined with locations of hospitals to show graphically how healthcare infrastructure corresponds with the virus's outbreak visually. The map shows teeming hotspots of infection that were thickly highlighted in a mixture of red and yellow dominantly enveloping most city centers into the rural areas and also across the Midwest down through the Southeast.

In fact, the dots representing hospitals are more concentrated in the urban regions, which corresponds with larger populations, while the dots in rural areas are sparser, reflecting fewer hospitals per capita. According to this map, one can find that a high infection rate often corresponds to highly concentrated markers of hospitals, possibly indicating that with enough medical facilities, the high volume of interpersonal contact in urban areas might be driving up the infection rates. On the other hand, rural areas with high infection rates but lower numbers of medical facilities have to fight the complications of late medial responses and how poor health infrastructure affects efficiency in the containment of outbreaks.

This trend emphasizes how meaningful the accessibility of healthcare is for pandemic control. Even the less densely populated areas, but with relatively fewer numbers of hospitals corresponding to the size of their populations, seem to struggle to keep infection rates at bay due to a lack of access to timely and adequate medical attention.

Transportation and Infection

SQL query:

```
SELECT
    c.state, c.county, a.name AS airport_name,
    AVG(c.jan1_2021) AS avg_cases_2021
FROM
    covid19_data c
JOIN
   airports a ON c.lat
   BETWEEN a.latitude - 0.5 AND a.latitude + 0.5 AND c.long
   BETWEEN a.longitude - 0.5 AND a.longitude + 0.5
GROUP BY
    c.state, c.county, a.name;
```

This query connects COVID-19 data with airport data by comparing geographic locations (using latitude and longitude) to explore the average number of COVID-19 cases in areas close to airports. This helps to understand the spread of disease in regions near airports, providing data support for disease prevention and control.

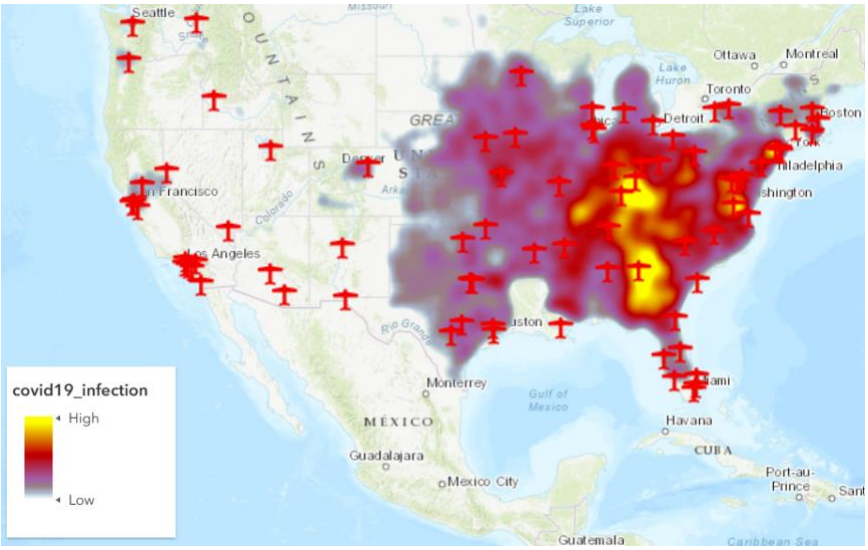


Figure 6: Transportation Hubs and COVID-19 Spread: A U.S. Heatmap Analysis

This analysis uses a map with overlaying major US airports over a heatmap of the infection rate of COVID-19 to attempt to show transport-related relationships to the transmission of the virus. Notice the map uses red cross indicators to point out each airport that serves over one million passengers a year. Starting at purple, the heating gets warmer and hotter up to red, to visualize increased infection intensities.

The densest red areas on the heatmap, reflecting the highest infection rates, often coincide with regions hosting major airport hubs, particularly in major metropolitan areas such as New York, Los Angeles, and Miami. These are areas characterized by high air traffic, while at the same time exhibiting elevated infection rates, and it may be that high mobility through transport hubs is promoting the spread of the virus.

While the most intense infection rates appear in urban centers with major airports, there are significant concentrations of infection in some rural communities that do not have nearby major airport facilities. This observation further implies that while airports act to significantly enhance the initiation of the spread in cities, the virus is independent of the airport factor and generally spreads into rural communities upon travel or from adjoining or nearby urban centers.

Conclusion

Our research mapped the distribution of COVID-19 infection rates across the United States and analyzed their association with a few important factors, namely population density, healthcare facilities, and proximity to major transportation hubs. These findings underlined the complex dynamics in pandemic spread and pointed out that besides expected difficulties in big cities, there is also significant vulnerability in rural areas. The findings of the study have proven that one cannot just trace high rates of infection in highly and densely populated areas, with some of the highest rates actually seen in low-population counties. The lack of facilities and infrastructure related to medical treatment in such places, alongside low mobility in the population, could have boosted the transmission rate and effects of the virus.

Discussion

Indeed, the data bear testament to the fact that upgrading health infrastructure and access in rural areas is, to a great extent, important in reducing the burden of COVID-19 or any other infectious disease outbreak among the populace. This refers to the need for medical facilities, staffing of the medical system accordingly, and the capacity and speed of response. Besides that, it underlines the role played by major transport hubs in the spread of the virus, representing nodes where public health interventions are needed more precisely with a view to flow control of potentially infected people.

These findings should be considered important for strategic enhancements in public health infrastructure and policy to better prepare against future health crises. Our analysis integrates geographic and demographic data and provides a sound framework for prioritizing resource investments and intervention strategies by public health officials. We therefore recommend a multidimensional response to the pandemic, integrating comprehensive knowledge from different sources for the various challenges global health crises pose.

Future Research

Further research could also extend this study to add real-time data tracking of the dynamics of virus spread over time. It would be interesting to explore the socioeconomic factors affecting healthcare accessibility and outcomes in more detail, with a view to providing insight into how interventions may most usefully be targeted. Research into behavioral patterns that drive personal and public compliance with health advisories across different demographic settings is needed.

This holistic approach helps in the understanding and realization of the pandemics-so important for building health-resilient systems and protecting current and future global populations throughout the outbreaks of infectious disease.

References

Centers for Medicare & Medicaid Services. October 2024. Hospital data dictionary. https://data.cms.gov/provider-data/sites/default/files/data_dictionaries/hospital/HOSPITAL_Data_Dictionary.pdf

U.S. Census Bureau. (2021). 2020 apportionment data tables. https://www.census.gov/data/tables/2020/dec/2020-apportionment-data.html

U.S. Census Bureau. (2024, September 26). Tiger Data Products Guide. Census.gov. https://www.census.gov/programs-surveys/geography/guidance/tiger-data-products-guide.html